

**DATA 3464: Fundamentals of Data Processing**

---

# **Bash and data cards**

Charlotte Curtis

March 12, 2026

# Topic overview

---

- Intro to bash and unix tools
- Data cards

## Resources used:

- [Data Science at the Command Line](#)
- [The Data Cards Playbook](#)

*Why these topics together? They both seem relevant to assignment 3*

# Why bash?

---

- Python is great, but it's not the only tool. Command line is better to:
  - Move and rename files
  - Take a peek at the first few lines of a giant csv
  - Find and replace text in a bunch of files
  - Fetch data from the web
- Bash and unix tools are more useful than their Windows counterparts
- [Git bash](#) lets us use bash on Windows

# Okay, but what is it?

---

- The [Bourne Again SHell](#) is a command line interface created in 1989
- A **shell** is a program to execute commands from the user
- The "Bourne shell" ( `sh` ) from the 1970s was standard on UNIX, bash adds to it
- Default shell for linux, (almost) macOS
- When you run these commands in Jupyter notebook, you're using bash!

```
%pip install -q some_package
```

# Basic bash

---

```
command_name --long_flag -l -o arg1 arg2
```

- Examples of common commands:
  - `ls` to list files in a directory
  - `cd` to change directories
  - `mv` to move or rename files
  - `head` to view the first few lines of a file
  - `grep` to search for text in files
  - `sed` to find and replace text in files
  - `curl` to fetch data from the web